Module 8

I²C — UVM+SV Verification

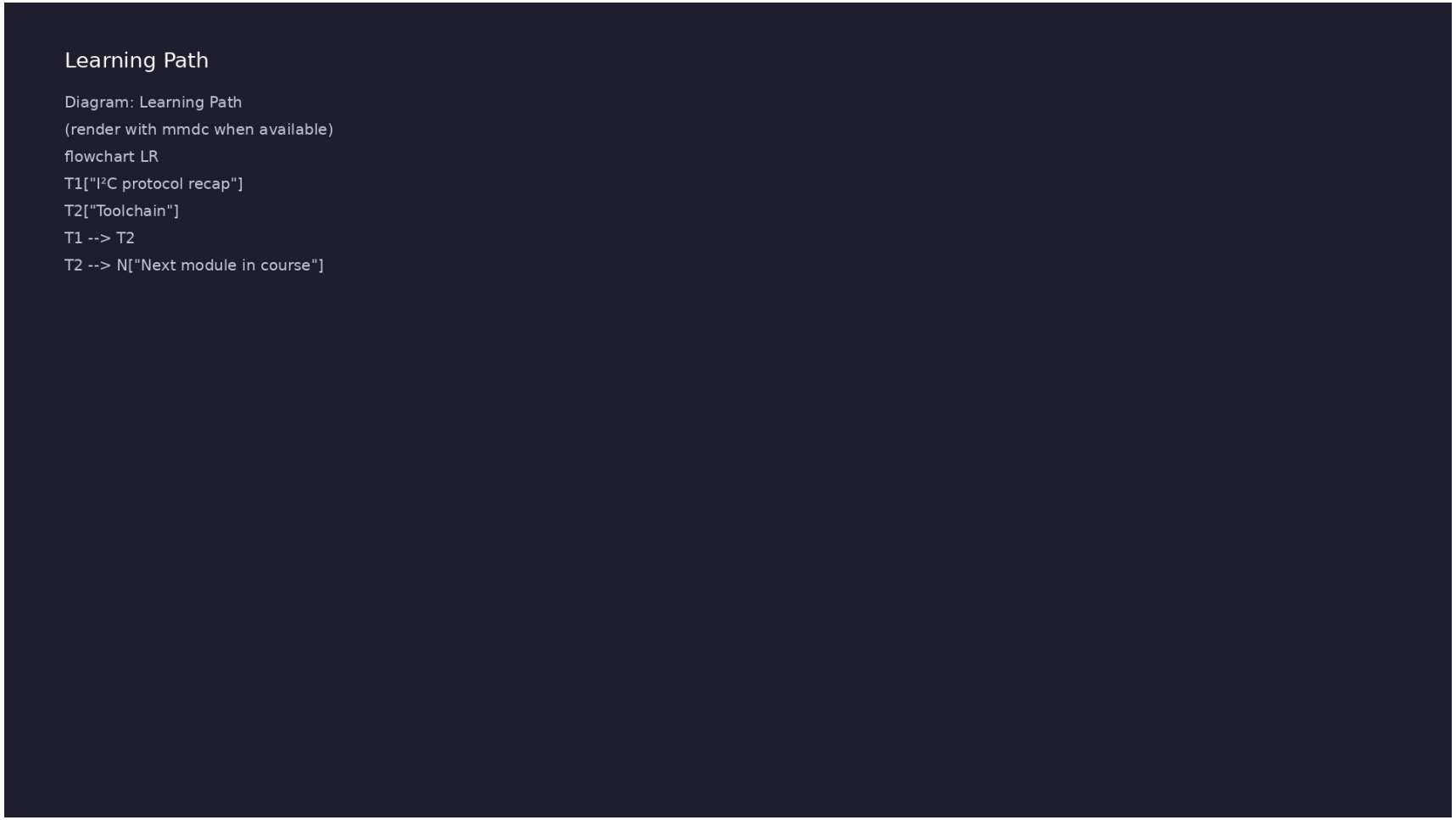
Learning objectives

- I²C UVM agent: I2cTransaction, I2cSequence, I2cDriver, I2cMonitor, I2cScoreboard.
- Interface: i2c_if (clk, rst_n, start, addr, data_in, scl, sda, clk_div_tick, busy, done) connects U…
- How to hook up a UVM monitor and scoreboard to a bus-oriented DUT.

Prerequisites

- Module 7 (I²C protocol + RTL + basic TB).
- Verilator, Make, C++ compiler, UVM (UVM_HOME or vendored UVM).

Learning path



Extend I²C verification to **UVM+SV** — I²C agent (transaction, sequence, driver, monitor, scoreboard); run on Verilator.

Overview

- Module 8 builds on Module 7 (I²C protocol + RTL + basic testbench):

How to learn this module

Suggested learning path

- Read the detailed I²C learning guide:

  I2C_LEARNING_GUIDE.md — what I²C is, how it works

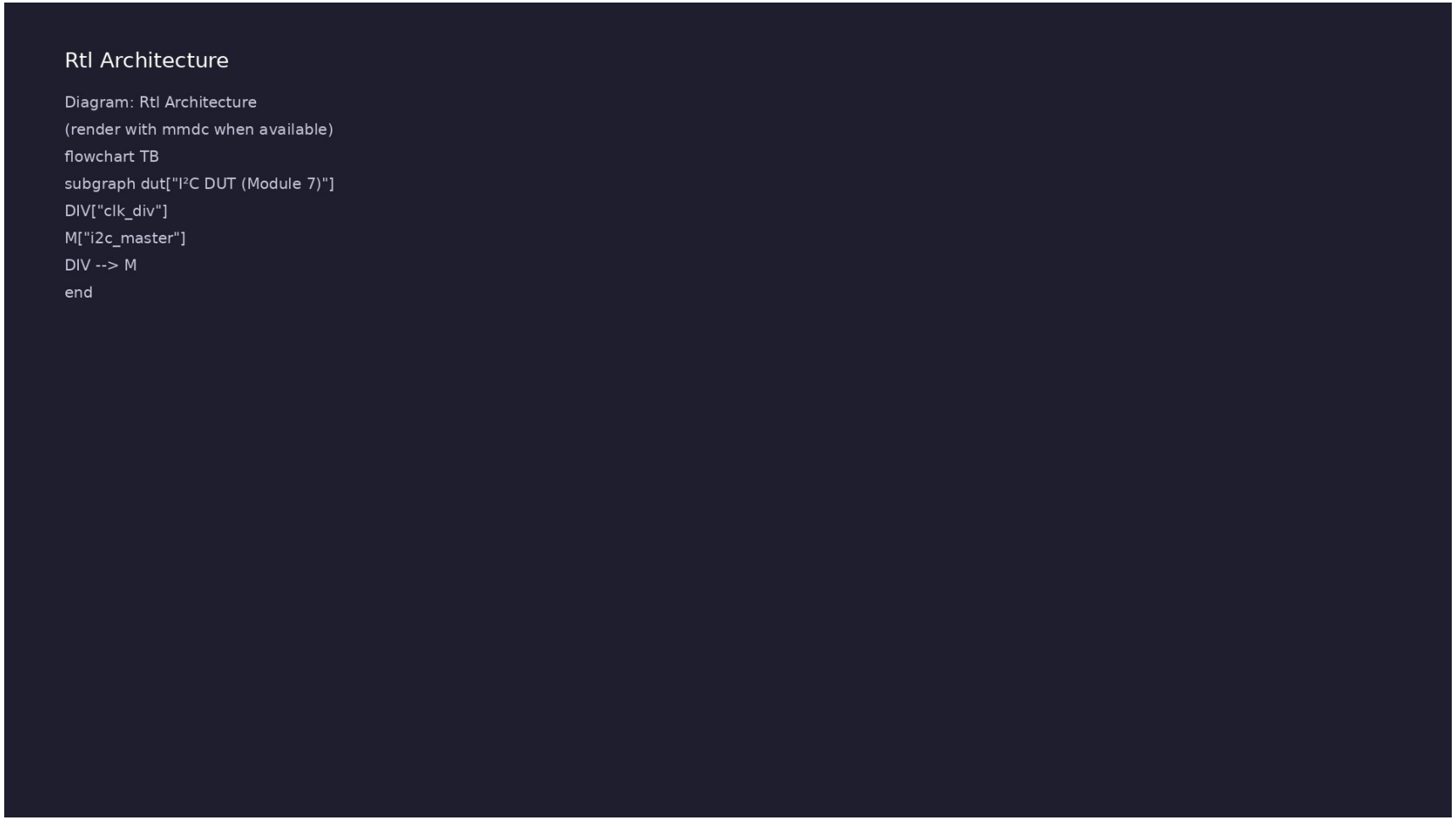
  (START/STOP…
- Read the protocols + UVM overview:

  LEARNING_GUIDE_PROTOCOLS_AND_UVM.md — Part B § 5.

  How UVM Maps t…

Design architecture

RTL / block architecture



Module 8: DUT / block hierarchy (see module8/examples/).

Verification environment architecture

Module 8: UVM layers, agents, and checkers for this phase.

1. I²C DUT (reused from Module 7)

- i2c_master + clk_div; i2c_if for UVM connection.
- Wire sequence: START → address+W → data → STOP.

2. UVM agent

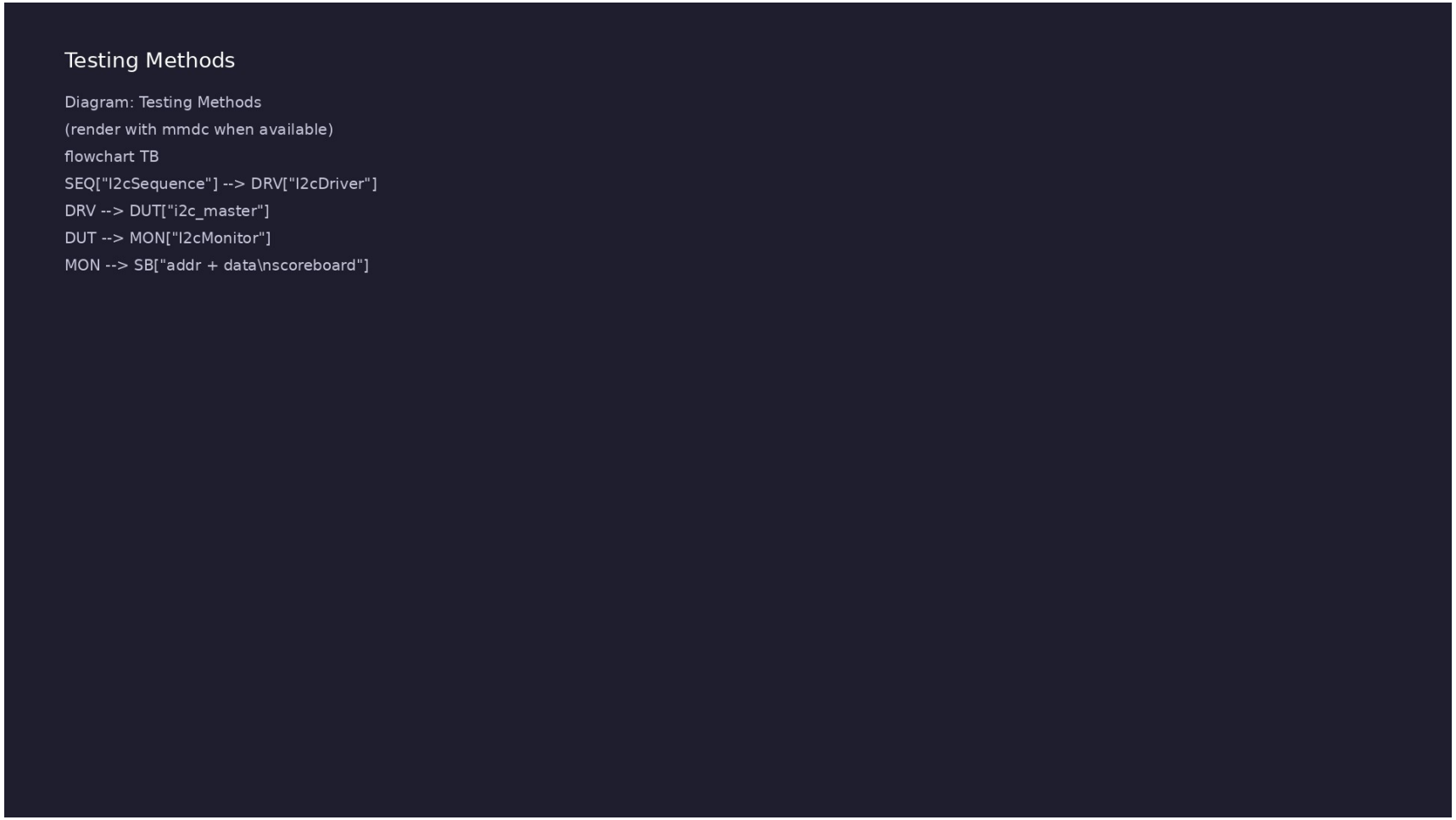
- I2cTransaction holds `addr` + `data`; monitor fills `observed_*`.
- I2cMonitor reimplements baseline bus sampling inside UVM.
- I2cScoreboard checks address and data phases independently.

3. Course capstone

- Compare UART (async), SPI (SCLK+CS), I²C (shared bus) — same UVM skeleton, different monitors.

Verification & testing methods

Testing and closure flow



Module 8: how tests, checks, and metrics close verification goals.

1. Two-phase scoreboard

- Separate checks for address+W byte and data byte — catches shifted or merged fields.

2. Reuse baseline monitor logic

- Module 7 inline monitor → Module 8 I2cMonitor for scalable tests.

3. Extensions

- ACK/NACK, reads, multi-master — [LEARNING_GUIDE § 8](LEARNING_GUIDE_PROTOCOLS_AND_UVM.md#8-i2c-uvm-...

Syllabus topics

1. I²C protocol recap

- START/STOP on SCL high; MSB-first address and data on SDA.

2. Toolchain

- `make SIM=verilator TEST=test_i2c_uvm`;
  `+UVM_TESTNAME=test_i2c_uvm`.

Hands-on examples

Module 8 self-check

```
$ ./scripts/module8.sh --help
```

Terminal: module8_check

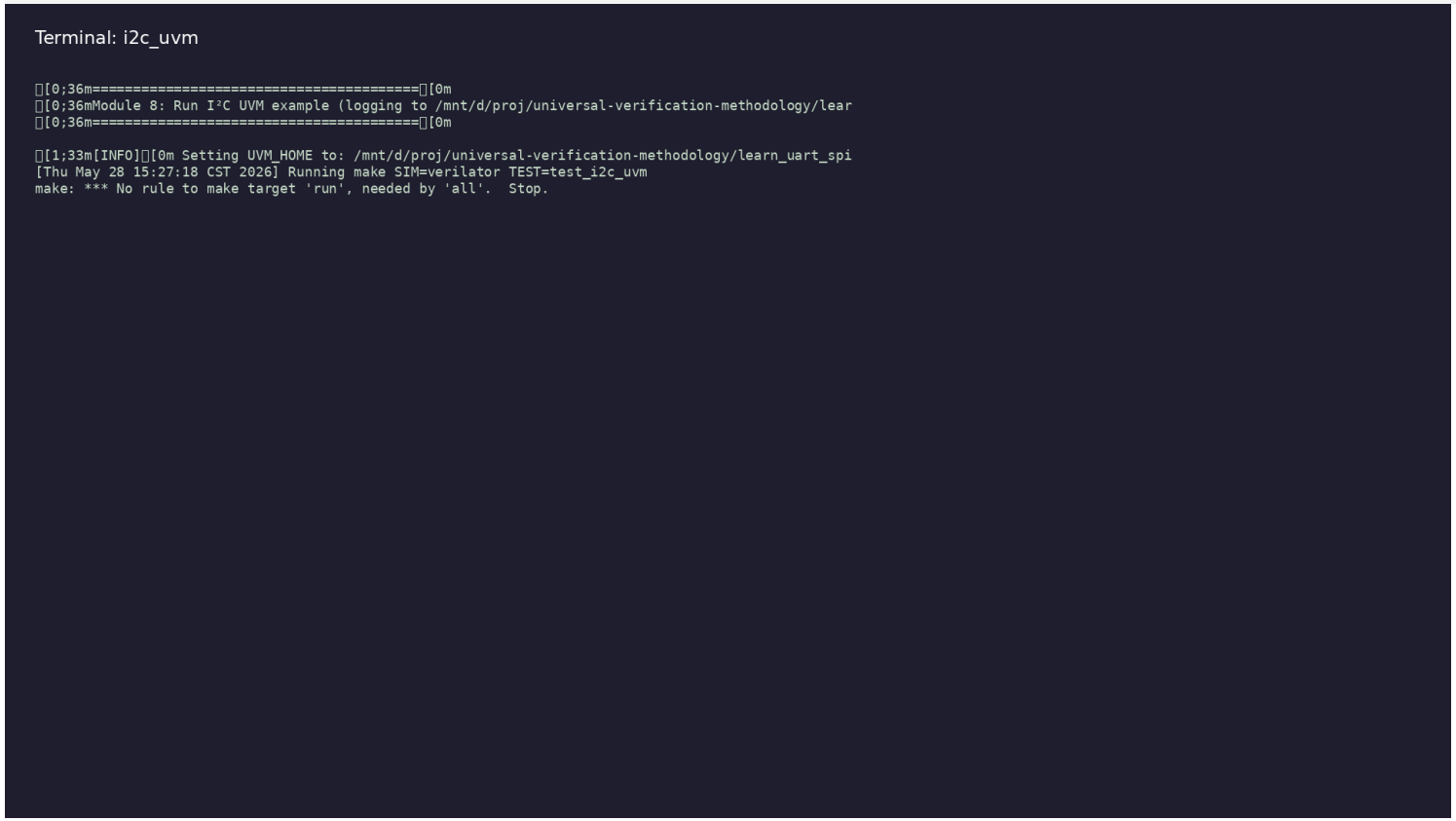
```
[0;36m========================================[0m
[0;36mModule 8: Demo commands (I²C UVM example)[0m
[0;36m========================================[0m

From repo root:

# 1) Environment sanity:
verilator --version
make --version
echo "$UVM_HOME"

# 2) Run the I²C UVM example (inside the example directory):
cd module8/examples/i2c_uvm
make SIM=verilator TEST=test_i2c_uvm

# 3) Inspect logs and waveforms:
ls obj_dir
# If a VCD is produced:
ls *.vcd

[1;33m[INFO][0m See module8/EXAMPLES.md and module8/examples/i2c_uvm/README.md for more details.
```

Example 1: I²C UVM

- UVM phases and reporting (DRIVER, MONITOR, SCOREBOARD).
- Scoreboard checks (expected vs observed address+data from the bus).
- Final scoreboard summary (matches and mismatches).

Demo: I²C UVM

```
$ ./scripts/module8.sh --run
```



Practice & assessment

What you should know

- Describe the I²C UVM agent and how I2cMonitor maps to SCL/SDA behavior.
- Explain two-phase scoreboard checking (address vs data).
- Run `i2c_uvm` and interpret UVM / scoreboard output.
- Compare verification architecture across UART, SPI, and I²C modules in this course.

Exercises

- Run i2c_uvm
- Trace one transaction
- Change the sequence
- Compare across protocols

Assessment checklist

- Can describe the I²C UVM agent and how it maps to I²C (addr+data, monitor on SCL/SDA).
- Can run i2c_uvm and interpret UVM output.
- Can compare UVM structure across UART, SPI, and I²C.

Summary & next steps

- Extend I²C verification to **UVM+SV** — I²C agent (transaction, sequence, driver, monitor, scoreboa…
- Complete module8/CHECKLIST.md
- Review module8/EXAMPLES.md and run each lab
- Next: Next module in course